Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**4**
Electrons (Ne)	**4**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**3**
Eigenstate idx	**4 (1-indexed: 5)**

Eigenstate properties

Energy E0	**-0.82842712**
Energy E1	**0.00000000**
Gap (E1-E0)	**0.828427**
<S^2>	**2.000000**
S	**1.0000**
<Sz>	**0.000000**
Multiplicity	**3.0**
Spin label	**Triplet (S=1)**

Double occupancy

<D> total	**0.146447**
<D> / site	**0.036612**

RDM and GME statistics for this run (3-body clusters)

Hilbert space dim : 2^3 = 8
Number of clusters : 0
Pure clusters : 0
Mixed clusters : 0
Clusters with GME (>0) : 0

Purity gamma = Tr(rho^2) : mean = — [min — , max —]
Von Neumann entropy : mean = — [min — , max —]
Rank of rho : mean = — [min — , max —]
C_GME (best estimate) : mean = — [min — , max —]

NOTE Analysis performed on eigenstate index 4 (state 5 in 1-indexed convention), not the ground state.

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Summary: All 3-Body RDM Clusters

System: L=4, Ne=4, U=4.0, t=1.0 | E_0=-0.828427

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state